Resit - exam with answers

Machine Learning (Vrije Universiteit Amsterdam)

Machine Learning 2023

Resit Exam

WITH ANSWERS

8 June 2023, 18:45–21:00

The questions start on the next page. **Please do not open the exam booklet until the scheduled starting time.**

Feel free to write on the exam booklet, and take it with you afterwards. The exam will be made available on Canvas later today, together with the correct answers. You can copy your final answers onto the booklet, so you can check how you did.

There are 40 questions, worth 1 point each.

To pass the examination, your total points for the exam plus those for the quizzes should be 52 or higher.

Rules:

- You are allowed to use a calculator or graphical calculator.
- You are *not* allowed to use your phone or smartphone.
- The exam is closed-book.
- You are allowed to use the formula sheet provided through Canvas.
- The formula sheet should not have any writing on it, except in the cheat sheet box. Anything inside the box should be written by hand.

Recall questions

1. The most important rule in machine learning is "never judge your performance on the training data." If we break this rule, what can happen as a consequence?
   A The loss surface no longer provides an informative gradient.
   B We get cost imbalance.
   C We end up choosing a model that overfits the training data.✓
   D We commit multiple testing.

2. We can use gradient descent to minimize a loss function. Which statement is **true**?
   A Gradient descent takes steps in the steepest descent direction, and this direction is the opposite of the gradient. ✓
   B Gradient descent takes steps in the steepest ascent direction, and this direction is the gradient.
   C Gradient descent takes steps in the steepest descent direction, and this direction is the gradient.
   D Gradient descent takes steps in the steepest ascent direction, and this direction is the opposite of the gradient.

3. In the slides, we get the advice that "sometimes your loss function should not be the same as your evaluation function." Why not?
   A The evaluation function may provide a smooth loss surface.
   B The loss surface resulting from the evaluation function may have a shape that makes searching for the minimum difficult.✓
   C The evaluation function may not be linear, and a loss function must be linear.
   D The evaluation function may not be a polynomial, and the loss function must be a polynomial.

4. There are many classification problems that are not linearly separable. Which trick often allows us to still separate the classes with a linear classifier?
   A Adding a regularizer.
   B Using least-squares-loss instead of log-loss.
   C Reducing the dimensionality with PCA.
   D Deriving new features from the existing ones.✓

5. We are training a classification model by gradient descent, and we want to figure out which learning rate to use, before comparing the model to other classifiers. We try five learning rate values, resulting in five different models. How do we choose among these five models?

   A We measure the accuracy of each model on the training set.
   B We measure the accuracy of each model on the validation set. ✓
   C We measure the accuracy of each model on the test set.
   D We measure the accuracy of each model on the full dataset.

6. Undersampling and oversampling are ways to deal with imbalanced classes. Which is **true**?

   A You oversample your majority class.
   B You undersample your minority class.
   C Undersampling leads to duplicate instances in your data.
   D Oversampling leads to duplicate instances in your data.✓

7. In the naive Bayes classifier, what does the word *naive* refer to?

   A The fact that prior probabilities are not taken into account.
   B The fact that the resulting classifier does not usually work very well in practice, and is only useful as a baseline.
   C The fact that the classifier makes an assumption of conditional independence that is not usually true.✓
   D The fact that the classifier makes an assumption of independence that is not usually true.

8. How does dropout help with the overfitting problem?

   A By propagating the gradient of the loss back down the network.
   B By randomly disabling nodes in a neural network, to eliminate solutions that require highly specific configurations.✓
   C By ensuring that the output distribution of a neural network is normally distributed if the input distribution is.
   D By converting the scalar backpropagation algorithm to work with tensors.

9. What is the *Markov assumption*? Does the LSTM make the Markov assumption?

   A That the probability of a word depends only on a fixed number of words before it. The LSTM makes a Markov assumption.
   B That the probability of a word depends only on a fixed number of words before it. The LSTM does not make a Markov assumption.✓
   C That it is more likely that a word should be forgotten than that it should be remembered. The LSTM makes a Markov assumption.
   D That it is more likely that a word should be forgotten than that it should be remembered. The LSTM does not make a Markov assumption.

10. When dealing with temporal data (i.e. data where each instance has a timestamp), it is important to keep the data ordered in time. What method allows us to evaluate models that do this, while also letting them retrain on new data once they've seen it?

    A Walk-forward validation✓
    B Cross-validation
    C Bootstrapping
    D Transductive learning

11. What is **mode collapse**?

    A When the loss surface of a generator network flattens out, so the gradient becomes zero.
    B When a network has a sampling step in the middle, so we cannot backpropagate down to the input.
    C When the distribution in the latent space is similar to a hypersphere, so we should interpolate along an arc instead of a line.
    D When a generator network outputs the mean of the data, instead of providing different samples with variation between samples.✓

12. By what method do variational autoencoders avoid mode collapse?

    A By training the "decoder" network through a discriminator.
    B By using a regularizer to steer the network toward the data average.
    C By feeding the discriminator network pairs of inputs.
    D By learning the latent representation of an instance through an "encoder" network.✓

13. What is the **cold start** problem?

    A The situation where a neural network is initialized so that its sigmoid activations are saturated, and it has no gradient to start learning with.
    B The situation in sequence-to-label learning where there is a long distance between the start of the sequence and the label, so that the model only learns from the end of the sequence.
    C The situation where we can sample from a generator network but we don't know which instance from the data to compare it to, because we don't have a mapping from the data to the latent space.
    D The situation where a new item (user, movie, etc.) is added to a recommender system, and we have no ratings to build an embedding from.✓

14. In some machine learning settings it is said that we must make a trade-off between *exploration* and *exploitation*.

    What do we mean by this?

    A That hyperparameter selection (exploration) uses computational resources that can also be used in training the model (exploitation).
    B That an insufficiently carefully trained model may be biased against minorities.
    C That a reinforcement learning algorithm needs to balance optimization of its expected reward with exploring to learn more about its environment.✓
    D The problem of balancing the loss function with the regularization terms in matrix factorization.

Combination questions

15. You want to search for a model in a discrete model space. Which search method is the **least** applicable?
    A Random search
    B Simulated annealing
    C Evolutionary methods
    D Gradient descent ✓

16. Jim is fitting a linear regression model to a dataset with a single feature x. He adds a second feature x^2. Which is **true**?
    A This may introduce a non-global optimum to the loss curve.
    B This allows Jim to model a nonlinear relation using linear regression.✓
    C This means that random search can no longer be used to fit the model.
    D This means that gradient descent can no longer be used to fit the model.

17. Which metric can you **not** compute from the confusion matrix?
    A Accuracy **B** True positive rate **C** Recall **D** ✓Area under the curve

18. One piece of advice we give in the course is that *the mean is sensitive to outliers*. What does this mean, and why is it the case?

A It means one instance can affect the mean much more than the others if its value is unusually large. This happens because the mean minimizes the absolute magnitude of the distances to all instances.
B It means one instance can affect the mean much more than the others if its value is unusually large. This happens because the mean minimizes the squares of the distances to all instances. ✓
C It means it can be a poor representation of the data if all instances are far from the average. This happens because the mean minimizes the absolute magnitude of the distances to all instances.
D It means it can be a poor representation of the data if all instances are far from the average. This happens because the mean minimizes the squares of the distances to all instances.

19. What happens if I build a two-layer neural network with no activation function on the units in the hidden layer?
    A The whole network becomes equivalent to a linear single layer network.✓
    B It becomes impossible to backpropagate a gradient to the first layer.
    C The optimization problem becomes non-convex.
    D We must use automatic differentiation to compute the gradient.

20. To make the backpropagation algorithm more broadly applicable, we introduced the *multivariate chain rule*. What was the main reason for this?
    A To allow us to use backpropagation on functions that are expressed in terms of computation graphs.
    B To allow us to apply backpropagation to computation graphs where the output depends on the input along multiple paths. ✓
    C To allow us to apply backpropagation to computation graphs where the output depends on the input only along a single path.
    D To make backpropagation a fully symbolic algorithm instead of a mixture of symbolic and numeric computation.

21. We have a logistic regression model for a binary classification problem, which outputs class probabilities q. We compare these to the true class probabilities p, which are always 1 for the correct class and 0 for the incorrect class. The slides mention two loss functions for this purpose: *log-loss* and *binary cross-entropy*. Which is **true**?

A Log-loss does not lead to a smooth loss landscape, so we approximate it by cross-entropy so that we can search for a good model using gradient descent.

B Cross-entropy loss does not lead to a smooth loss landscape, so we approximate it by log probability so that we can search for a good model using gradient descent.

C Log-loss is equal to the binary cross-entropy $H(p, q)$. ✓

D Log-loss is equal to the binary cross-entropy $H(q, p)$.

22. In a maximum-margin loss, which is **not** an advantage of using *soft margins*?

A It allows some points to fall inside the margin.

B It allows some points to be misclassified.

C It allows us to use gradient descent for optimization, which a hard margin doesn't. ✓

D It finds better decision boundaries even for linearly-separable classes.

23. Can we use Gaussian Mixture models (GMMs) for classification?
A Yes. One way is to split the data by class and to fit a separate GMM to the points for each class. ✓
B Yes. A hard margin support vector machine is technically equivalent to a GMM-based classifier.
C No. GMMs are density estimation models.
D No. GMMs are regression models.

24. What connects generative adversarial networks and variational autoencoders?
A They both consist of a generator network that generates random samples and a discriminator network that tells real from fake samples.
B They both train a generator network consisting of a random input fed to a deterministic neural network. ✓
C They both require reparametrization to allow backpropagation through a sampling step.
D They both use cycle-consistency loss to deal with unpaired images in the training data.

25. What problem, if it exists for a single model, **cannot** be solved by training an ensemble of those models?
A High training time.✓
B High bias.
C High variance.
D High overfitting.

Application questions

We have the following training set:

	x_1	x_2	label
a	1	3	Pos
b	2	2	Pos
c	3	1	Neg
d	6	4	Neg
e	5	5	Pos
f	4	6	Pos
g	7	7	Neg

For the following questions, it helps to draw the data and the classification boundary in feature space.

We use a linear classifier defined by

$$c(x_1, x_2) = \begin{cases} \text{Pos} & \text{if } -0 \cdot x_1 + x_2 > -2 \\ \text{Neg} & \text{otherwise.} \end{cases}$$

26. If we turn c into a *ranking* classifier, how does it rank the points, from most Negative to most Positive?
    A c b a d e f g ✓
    B g f e d a b c
    C g d e f c b a
    D a b c f e d g
27. How many ranking errors does the classifier make?
    A 3
    B 4
    C 5
    D 6✓
    If you answered 3, make sure you understand the difference between a *ranking error* and a *classification error*. Classification errors as instances that the classifier classifies incorrectly (in this case the instances b, e, and f) and ranking errors are *pairs* of positive and negative instances that a ranking classifier ranks the wrong way around. In this case (c, e) and (c,f)
28. If we draw a coverage matrix (as done in the slides), what proportion of the cells will be red?

    A $\frac{1}{2}$✓ **B** $\frac{1}{3}$ **C** $\frac{1}{4}$ **D** $\frac{1}{6}$
    The size of the coverage matrix is the number of positives times the number of negatives ($4 \times 3 = 12$). The number of red cells is the number of ranking errors (2) so the proportion of red cells is $6/12 = 1/2$

We will use the backpropagation algorithm to find the derivative of the function

$$f(x) = \sin\left(3x^2 + \cos x\right)$$

with respect to x.

First, we break the function up into modules:

$$a = 3x^2$$
$$b = \cos x$$
$$c = a + b$$
$$f = \sin c$$

29. What should be in the place of the dots?
    A $b = \cos x,\ \ c = a + b$ ✓
    B $b = \cos x,\ \ c = \sin b$
    C $b = \cos a,\ \ c = a + b$
    D $b = \cos a,\ \ c = \sin b$

30. In terms of the *local* derivatives. Which is the correct expression for the derivative $\frac{\partial f}{\partial x}$?

A $\cos(c)\ 6x - \sin x$

B $\cos(c)\ 6x - \sin a$

C $\cos(c)\ (6x - \sin a)$

D $\cos(c)\ (6x - \sin x)$✓

We have trained a linear support vector machine, and found the following weights and bias:

$$w = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \qquad b = 0. \tag{1}$$

31. Which of the following points are *support vectors* for the trained model?

A $y_1 = 1, \mathbf{x}_1 = (1, 1)$ $\quad y_2 = -1, \mathbf{x}_2 = (-1, -1)$

B $y_1 = 1, \mathbf{x}_1 = (-1, -1)$ $\quad y_2 = -1, \mathbf{x}_2 = (1, 1)$

C $y_1 = 1, \mathbf{x}_1 = (-1, 1)$ $\quad y_2 = -1, \mathbf{x}_2 = (1, -1)$

D $y_1 = 1, \mathbf{x}_1 = (1, -1)$ $\quad y_2 = -1, \mathbf{x}_2 = (-1, 1)$✓

32. Consider two points: $\mathbf{x}_1 = (1, 1)$ and $\mathbf{x}_2 = (0, -1)$. How are these two points classified?

A Both positive.

B $\mathbf{x}_1$ is positive, $\mathbf{x}_2$ is negative.✓

C $\mathbf{x}_1$ is negative, $\mathbf{x}_2$ is positive.

D Both negative.

We are given a dataset of email, labeled spam and ham. The total number of words in the spam and ham datasets is is 30 000 and 60 000, respectively.

On this dataset, we want to train a *first-order Markov model* as a basis for a Bayes classifier to detect spam.

The following table shows the frequency of several bigrams and unigrams in the dataset.

uni- and bigrams	frequency spam	frequency ham
about	3 000	3 000
that	5 000	5 000
meeting	1 000	2 000
tomorrow	1 000	2 000
about that	100	200
that meeting	100	200
meeting tomorrow	100	200

We have two *priors* on how likely an email is to be spam. **Prior 1** says that an email is as likely to be spam as ham. **Prior 2** says that a spam email is four times as likely as a ham email.

Consider the following email:

email: | about that meeting tomorrow |

33. How is the email classified?
    A With both priors as ham.
    B With both priors as spam.
    C Under prior 1 as spam, under prior 2 as ham.
    D Under prior 1 as ham, under prior 2 as spam.✓

34. Under prior 2, what is the probability that the email is spam?
    A $\frac{1}{3}$ **B** $\frac{2}{3}$✓ **C** $\frac{4}{7}$ **D** $\frac{3}{7}$

35. Under prior 1, what is the *probability ratio* of ham over spam for the email?
    A 2✓ **B** $\frac{1}{2}$ **C** $\frac{3}{15}$ **D** 5

Consider the following function:

$$f(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T W \mathbf{y}$$

where $\mathbf{x}$ and $\mathbf{y}$ are vectors and W is a matrix. Note that this makes f a function from two vectors to a scalar.

This function is part of a larger computation graph, resulting in a scalar loss l. We want to implement this computation in an automatic differentiation (AD) system, as discussed in the lectures.

36. Work out the local scalar derivative $\partial f / \partial x_i$. Which is the correct solution?
    A $\partial f / \partial x_i = \sum_j W_{ij} y_j$✓
    B $\partial f / \partial x_i = \sum_j W_{ji} y_j$
    C $\partial f / \partial x_i = \sum_j x_i W_{ij}$
    D $\partial f / \partial x_i = \sum_j x_i W_{ji}$

The AD system will give us a scalar $f^\nabla = \frac{\partial l}{\partial f}$. Using f^∇, we want to efficiently compute a vector $\mathbf{x}^\nabla$ such that $x_i^\nabla = \frac{\partial l}{\partial x_i}$.

37. Which operation computes this vector for us?
    A $\mathbf{x}^\nabla = f^\nabla \cdot \mathbf{x} W$
    B $\mathbf{x}^\nabla = f^\nabla \cdot \mathbf{x} W^T$
    C $\mathbf{x}^\nabla = f^\nabla \cdot W \mathbf{y}$ ✓
    D $\mathbf{x}^\nabla = f^\nabla \cdot W^T \mathbf{y}$

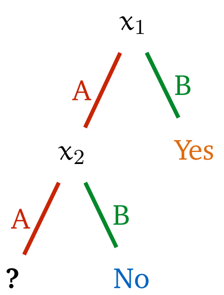
Consider the following task. The aim is to predict the class y from the binary features x_1, x_2, x_3 and x_4.

x_1	x_2	x_3	x_4	y
A	A	A	A	No
A	B	A	A	No
A	A	A	A	No
A	A	A	A	No
B	B	A	A	No
B	B	B	A	No
A	B	A	B	Yes
A	A	B	B	Yes
B	A	B	B	Yes
B	A	B	B	Yes
B	B	B	B	Yes
B	B	B	B	Yes

38. In standard decision tree learning (as explained in the lectures), without pruning, which would be the first feature chosen for a split?
A x_1 **B** x_2 **C** x_3 **D** x_4✓

39. If we *remove* that feature from the data, which would be chosen instead?
A x_1 **B** x_2 **C** x_3✓ **D** x_4

Consider the following (partial) decision tree:



Note that this is just an arbitrary tree, not necessarily corresponding to the answers in the previous questions.

We can place either x_3 or x_4 on the open node (indicated by the question mark).

40. At this point, what is the information gain for x_3?
A 0 **B** $-\frac{1}{4}\log_2(3) + 2$ **C** $-\frac{3}{4}\log_2(3) + 2$✓ **D** 1

Thank you for your effort. Please check that you've put **your name and student number** on the answer sheet.